

Conjecture

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1 Reformulation of CVP

Definition 1.1. (*Problem X*)

Let $\mathbf{y} = (y_1, y_2, \dots, y_n) \in \mathbb{Z}^n$. Consider a non-homogeneous system:

$$\psi_k(\mathbf{y}) = c_k + \sum_j^n b_{kj}y_j, \quad 1 \leq k \leq t$$

Consider a homogeneous system:

$$\phi_i(\mathbf{y}) = \sum_j^n a_{ij}y_j, \quad 1 \leq i \leq m$$

The problem is to optimize the following:

$$\min_{\mathbf{y} \in \mathbb{Z}^n} \left(\sum_{i=1}^m |\phi_i(\mathbf{y})|^2 \right)^{1/2}$$

subject to $\psi_k(\mathbf{y}) = 0 \quad \forall \quad 1 \leq k \leq t$

Remark 1. We don't need $\phi_i(\cdot)$ per se to define Problem X. But we are using this because we might need it in our reduction from some other problem to Problem X.

1.1 Reduction from Problem X to CVP

Let us assume Problem X has a particular solution $\mathbf{y}^0$ with integer coordinates. That is, we are saying the non-homogeneous system $\psi_k(\mathbf{y}) = 0$ has atleast one integral solution. Consider the homogeneous system corresponding to $\psi_k(\cdot)$:

$$\begin{aligned} \theta_k(\mathbf{y}) &= \psi_k(\mathbf{y}) - c_k \\ \Rightarrow \theta_k(\mathbf{y}) &= \sum_j b_{kj}y_j, \\ &\quad \forall \quad 1 \leq k \leq t \end{aligned}$$

$$\begin{aligned} \psi_k(\mathbf{y}^0) = 0 &\Rightarrow c_k + \sum_j b_{kj}\mathbf{y}_j^0 = 0 \\ &\Rightarrow c_k = - \sum_j b_{kj}\mathbf{y}_j^0 \end{aligned} \tag{1}$$

Let $\mathcal{B}$ be the lattice corresponding to integral solutions of $\theta_k(\cdot) = 0$. Further let $\mathbf{y}^1$ be any arbitrary solution to the system $\psi_k(\cdot) = 0$.

$$\begin{aligned}
& \Rightarrow \psi_k(\mathbf{y}^1) = 0 \\
& \Rightarrow c_k + \sum_j b_{kj} \mathbf{y}_j^1 = 0 \\
& \Rightarrow - \sum_j b_{kj} \mathbf{y}_j^0 + \sum_j b_{kj} \mathbf{y}_j^1 = 0 \text{ [Using (1)]} \\
& \Rightarrow \sum_j b_{kj} (\mathbf{y}_j^1 - \mathbf{y}_j^0) = 0 \\
& \Rightarrow \theta_k(\mathbf{y}^1 - \mathbf{y}^0) = 0 \\
& \Rightarrow \mathbf{y}^1 - \mathbf{y}^0 \in \text{Lattice } \mathcal{B}
\end{aligned} \tag{2}$$

Thus, any solution of the system $\psi_k(\mathbf{y}) = 0$ is of the form $\mathbf{y}^0 - l$ where l is an element of the Lattice $\mathcal{B}$. (For solution $\mathbf{y}'$ of $\psi_k(\cdot) = 0$, $l = \mathbf{y}^0 - \mathbf{y}'$)

Problem X corresponds to minimizing $\|\phi \mathbf{y}\|$ over $\mathbf{y} \in \mathbb{Z}^n$ subject to $\psi \mathbf{y} = 0$. This is same as minimizing $\|\phi(\mathbf{y}^0 - l)\|$ over $l \in \mathcal{B}$. This is same as minimizing the distance of point $\phi \mathbf{y}^0$ from the Lattice $\phi \mathcal{B}$. Hence, we get an instance of CVP

Remark 2. $\mathbf{y}^0$ is, in general, not a point on Lattice $\mathcal{B}$.

Remark 3. We don't care about the case when the system $\psi_k(\cdot) = 0$ has no integral solutions because even Problem X is vacuous in that case.

2 Label Cover to Problem X reduction

Definition 2.1. (*Label Cover*)

$G = (V, W, E)$ is a bipartite graph. V : left vertices, W : right vertices, E : edges. The graph is left regular i.e., all vertices in V have the same degree. $n = |V|$ and $m = |W|$.

We have to assign one **label** to every vertex. Vertices in V receive label from set $[R]$ and vertices from W receive label from set $[S]$. A labelling A is a map. $A : V \mapsto [R], W \mapsto [S]$.

The labelling is supposed to satisfy constraints given by maps $\pi_{v,w} : [R] \mapsto [S]$. There is one such map for every edge $(v, w) \in E$

Remark 4. There are multiple maps, one for each pair of vertices (v, w) . It is like if (v, w) are supposed to have labels from $\{(r_1, s_1), (r_2, s_2), \dots, (r_\mu, s_\mu)\}$.

A labelling A **satisfies** an edge (v, w) , if

$$\pi_{v,w}(A(v)) = A(w)$$

Remark 5. We require the maps $\pi_{v,w}$ to be functions (or single-valued maps);

Remark 6. In this setting, a labelling must assign each vertex **exactly one** label. Nothing is said about whether 2 vertices can have same label. It is allowed most likely.

The optimum $OPT(LC)$ of the instance is defined to be the maximum fraction of edges satisfied by any labelling.

We assume $n \geq m$, and $R \geq S$. (Left side is larger)

Definition 2.2. (GapCVP)

γ – GapCVP is a promise problem $(\mathcal{L}(\mathbf{B}), \mathbf{t}, r)$ whose YES instances satisfy $dist(\mathbf{t}, \mathcal{L}(\mathbf{B})) \leq r$, and NO instances satisfy $dist(\mathbf{t}, \mathcal{L}(\mathbf{B})) \geq \gamma r$.

Theorem 2.3. [*Kho09*] There exists an absolute constant $\beta > 0$ such that for every integer $R \geq 7$, there is a reduction from SAT instance ϕ to Label Cover instance $LC(G(V, W, E), n, m, \{\pi_{v,w}\}, [R], [S])$ with the following property:

- The YES instances of SAT map to label cover instances with $OPT(LC) = 1$.
- The NO instances map to label cover instances with $OPT(LC) \leq R^{-\beta}$.

The running time of the reduction and the size of the label cover instance are bounded by $|\phi|^{O(\log R)}$.

Remark 7. Absolute constant means β is independent of R .

Remark 8. The only freedom we have is in choosing R . Note that the large integer R is same used in left-label Universe $[R]$.

2.1 Reduction

Let $LC(G(V, W, E), n, m, \{\pi_{v,w}\}, [R], [S])$ be the instance of Label Cover given by *Thm 2.3*(2.3).

Remark 9. This is in essence an instance of GapLC.

We describe a reduction from this instance to an instance of $\gamma - \text{GapCVP}$ with gap $\gamma = \frac{1}{20}R^{\beta/2}$.
Remark 10. Note that we have freedom with R . So we can choose appropriate gap by adjusting R . Instead of giving a direct reduction to GapCVP , we give a reduction to Gap variant of Problem X.

2.1.1 Problem X construction

The set of integer valued variables ($\mathbf{y}$ in Problem X) is:

$$Y := \{y_{v,j} \mid v \in V, j \in [R]\} \cup \{z_{w,i} \mid w \in W, i \in [S]\}$$

The function to be minimized (optimization problem in Problem X) is:

$$OBJ := \left(m \cdot \sum_{v \in V, j \in [R]} y_{v,j}^2 + n \cdot \sum_{w \in W, i \in [S]} z_{w,i}^2 \right)^{1/2}$$

The *affine* constraints ($\psi_k(\cdot) = 0$ in Problem X) are:

$$\forall v \in V, \sum_{j \in [R]} y_{v,j} = 1 \quad (3)$$

$$\forall w \in W, \sum_{i \in [S]} z_{w,i} = 1 \quad (4)$$

$$\forall (v, w) \in E, \forall i \in [S], z_{w,i} = \sum_{j \in [R]: \pi_{v,w}(j)=i} y_{v,j} \quad (5)$$

2.1.2 Remarks

Remark 11. There are $n \cdot R + m \cdot S$ variables.

Remark 12. These variables can take all integer values, not just 0 and 1.

Remark 13. These variables don't represent anything, in general. In particular, these variables are not to be seen as Indicator variables.

Remark 14. In case of a YES instance of GapLC , there exists a 0,1 assignment of these variables which would then correspond to an optimum labeling ($OPT(LC) = 1$ labelling). The same assignment corresponds to optimum of Problem X as well. ($OBJ = \sqrt{2mn}$)

Remark 15. In case of NO instance of GapLC , we argue that every integer assignment of these variables results in OBJ value much greater than $\sqrt{2mn}$.

Remark 16. To figure out: Why m and n weights attached to summations in OBJ . Maybe to get an expression like $\sqrt{2mn}$ that we can easily handle. But are m and n necessary for the reduction?

Remark 17. If we consider only 0,1 assignments, we get a 1-1 correspondence of assignments with valid labellings.

Remark 18. If we consider only 0,1 assignments to variables, then YES instance will have a 0,1 assignment that satisfies all constraints (3)-(5). This is because YES instance of GapLC has $OPT(LC) = 1$.

Remark 19. In case of NO instance of GapLC, no labelling can satisfy all the edges. Thus, no 0,1 assignment can satisfy all constraints (3)-(5).

Remark 20. We have 2 options of dealing with the issue of NO instances:

- Allow some constraints amongst (3)-(5) to be unsatisfied.
- Expand the solution space to allow the variables to take integer values other than 0,1 as well. But the variables must satisfy all the constraints (3)-(5).

First option doesn't give us good traction on the problem. There are too many degrees of freedom. What constraints are non-negotiable, what are negotiable? Do they capture the essence of Label Cover problem? These questions don't give positive answers.

One thing we can try is: “What is OBJ for an assignment that satisfies maximum number of constraints (3)-(5).” This leads to a dead-end. What constraints are negotiable? If (3),(4) are negotiable, then the assignment no longer represents a labelling with exactly one label to each vertex. On the other hand, if (3),(4) are non-negotiable, then OBJ is always equal to $\sqrt{2mn}$. This destroys the *gap* between YES/NO instances.

Hence, we consider the second option. We allow the variables to take integer values other than 0,1 as well. This relaxation would give assignments that might not represent a Label Cover instance on face value. But we prove that this relaxation preserves the YES/NO instances. YES instances still have $OBJ \leq \sqrt{2mn}$ and NO instances still have $OBJ \geq 0.1 \cdot R^{\beta/2} \sqrt{mn}$. (Recall that we have freedom with R . Thus, we can choose R big enough to get desired gap.)

Remark 21. We can have the case that no integer assignment to the variables satisfies all constraints (3)-(5). This is a vacuous case for which we can adequately define $OBJ := +\infty$.

Remark 22. To figure out: The reduction in reverse direction from this particular variant of Problem X to GapLC seems unlikely (atleast direct reduction seems infeasible). A particular instance of Problem X might have optimum OBJ corresponding to some *non-binary* integer assignments to variables. Such assignments don't have a direct correspondence in the Label Cover Problem in present setting.

Remark 23. Let us go back to first principles. A reduction from problem A to problem B means that if we have a subroutine to solve Problem B, then we can use it to solve Problem A. In a sense an arbitrary instance of Problem A can be converted to an instance of Problem B, such that the YES instance of Problem A map to YES instance of Problem B, and NO instances of Problem A map to NO instances of Problem B.

Now Problem A is Gap-Label Cover, where gap means YES instances have a labelling that covers all edges and NO instances cannot have more than $R^{-\beta}$ fraction of edges covered for any labelling. Here, the YES/NO instances are preserved, and the gap is transformed from $R^{-\beta}$ to $\frac{1}{20}R^{\beta/2}$. β is an absolute constant and $R \geq 7$ is an integer that can be chosen as large as desired.

2.1.3 YES Case

We prove that if the Label Cover has a labelling that satisfies all edges, i.e., $OPT(LC) = 1$, then there is an integer assignment to the variables in Y such that $OBJ \leq \sqrt{2mn}$.

Let $A : V \mapsto [R], A : W \mapsto [S]$ be such a labelling. Define:

$$y_{v,j} = \begin{cases} 1 & \text{if } j = A(v) \\ 0 & \text{otherwise} \end{cases}$$

$$z_{w,i} = \begin{cases} 1 & \text{if } i = A(w) \\ 0 & \text{otherwise} \end{cases}$$

Since there is exactly one label for each vertex, for each $v \in V$, exactly one $y_{v,j} = 1$ and rest all are 0. Similarly, for each $w \in W$, exactly one $z_{w,i} = 1$ and rest all are 0. Therefore,

$$OBJ = \sqrt{m \cdot |V| + n \cdot |W|} = \sqrt{2mn}$$

Exactly one label condition for each vertex ensures (3) and (4) are satisfied. Now, we show that all constraints in (5) are satisfied. Consider any $(v, w) \in E$ and $i \in [S]$. We will show that

$$z_{w,i} = \sum_{j \in [R] : \pi_{v,w}(j) = i} y_{v,j} \quad (6)$$

Let $i_0 = A(w)$ and $j_0 = A(v)$. This implies $z_{w,i_0} = 1$, $y_{v,j_0} = 1$. Since the labelling satisfies all edges, it satisfies edge $(v, w) \in E$ in particular as well. Thus, we have $\pi_{v,w}(j_0) = i_0$. Moreover, since each vertex has exactly one label, we have $\forall i \neq i_0, z_{w,i} = 0$ and $\forall j \neq j_0, y_{v,j} = 0$. Hence, (6) holds for $i = i_0$.

For $i \neq i_0$, LHS = 0. Since $\pi_{v,w}$ is a function, therefore $\pi_{v,w}(j_0)$ gives atmost 1 output. (exactly one here, $\pi_{v,w}(j_0) = i_0$) Thus, for $i \neq i_0$, the summation does not include y_{v,j_0} . Since $\forall j \neq j_0, y_{v,j} = 0$, we have RHS = 0 as well. Hence, (6) holds for $i \neq i_0$ too.

Thus, in YES case, we explicitly give an assignment satisfying (3)-(5).

2.1.4 NO case

Let $\alpha = R^{-\beta}$. Since we have freedom with the choice of R , we can choose R large enough to ensure $\alpha < 0.1$. Hereafter we assume $\alpha < 0.1$.

We prove that if the Label Cover instance has no labelling that satisfies even α fraction of its edges, then for any integer assignment to variables in Y (*any means positive, negative, non-binary all allowed*) that satisfies *all* constraints (3)-(5), one must have:

$$OBJ \geq 0.1 \sqrt{mn/\alpha}$$

Since we are considering Label Cover instances given by (2.3), NO instances will have $OPT(LC) \leq R^{-\beta}$. Thus, if the aforementioned result is proven, we will have:

$$OBJ \geq 0.1 \cdot R^{\beta/2} \sqrt{mn}$$

Therefore the gap between YES and NO cases is:

$$\frac{0.1 \cdot R^{\beta/2} \sqrt{mn}}{\sqrt{2mn}} \geq \frac{1}{20} R^{\beta/2} \quad \text{as claimed}$$

Proof.

Consider any *integer* assignment A to variables in Y that satisfies *all* constraints (3)-(5). Define sets $T_v \subseteq [R]$, $T_w \subseteq [S]$ as:

$$T_v := \{j \in [R] \mid y_{v,j} \neq 0\}, \quad T_w := \{i \in [S] \mid z_{w,i} \neq 0\}$$

Due to constraints (3) and (4), the sets T_v, T_w are non-empty for all $v \in V$, $w \in W$.

Remark 24. In essence, the sets T_v, T_w represent what variables got assigned non-zero values corresponding to each vertex.

Remark 25. Note that these variables don't correspond to any labelling of vertices per se. We will later show that any labelling can be visualised in this setting. This assignment problem, infact, is a superset of all possible labellings. We are proving that even for this superset, we always get $OBJ \geq 0.1\sqrt{mn/\alpha}$. Then it implies, for all valid labellings as well, we get $OBJ \geq 0.1\sqrt{mn/\alpha}$.

Remark 26. If no integer assignment to the variables in Y satisfies all constraints (3)-(5), then $OBJ := +\infty$. This correctly corresponds to **NO** instance of Label Cover.

Lemma 2.4. *For any $(v, w) \in E$ and $i \in T_w$, there exists $j^* \in T_v$ such that $\pi_{v,w}(j^*) = i$.*

Proof.

Consider the constraint (5) $z_{w,i} = \sum_{j \in [R]: \pi_{v,w}(j)=i} y_{v,j}$. Since $i \in T_w$, we have $z_{w,i} \neq 0$. Now, $z_{w,i} \neq 0$, in turn, implies atleast one of the variables on the right side of the constraint (5) must be non-zero. Let y_{v,j^*} be such a non-zero variable. Since $y_{v,j^*} \neq 0$, we have $j^* \in T_v$. Thus, we have $j^* \in T_v$ and $\pi_{v,w}(j^*) = i$, as desired.

Remark 27. The Lemma (2.4) in essence says, for each edge, if the right endpoint gets some non-zero assignment to a variable, then there has to be a corresponding variable under $\pi_{v,w}$ getting non-zero assignment on the left endpoint as well. In other words, for any edge, right endpoint gets no non-zero assignment without a corresponding non-zero assignment on left endpoint. This is basically a reformulation of Constraint (5) in new setting.

We consider 2 cases depending on whether the typical size of sets T_v is *small* or *large*. Let

$$V^* = \{v \in V \mid |T_v| \geq 0.1/\alpha\}.$$

Remark 28. Since we assume $\alpha < 0.1$, we have $0.1/\alpha > 1$. Thus, the set V^* is non-trivial. (Recall that each set T_v is non-empty, so $|T_v| \geq 1$ would have been trivially true for each v .)

Case 1 $|V^*| \geq 0.1|V| = 0.1 \cdot n$. We have

$$\begin{aligned}
OBJ &= \left(m \cdot \sum_{v \in V, j \in [R]} y_{v,j}^2 + n \cdot \sum_{w \in W, i \in [S]} z_{w,i}^2 \right)^{1/2} \\
&\geq \left(m \cdot \sum_{v \in V, j \in [R]} y_{v,j}^2 \right)^{1/2} \\
&\geq \left(m \cdot \sum_{v \in V^*, j : y_{v,j} \neq 0} 1 \right)^{1/2} \\
&= \left(m \cdot \sum_{v \in V^*} |T_v| \right)^{1/2} \\
&\geq \sqrt{m \cdot |V^*| \cdot 0.1 / \alpha} \\
&\geq 0.1 \sqrt{mn / \alpha}
\end{aligned} \tag{7}$$

Thus, we get $OBJ \geq 0.1 \sqrt{mn / \alpha}$, as desired.

Case 2 $|V^*| < 0.1|V| = 0.1 \cdot n$.

Ignore the edges (v, w) that are incident on V^* . (V^* is a set consisting of left vertices. We are ignoring the edges that have left endpoint as a vertex from V^*) Since, the graph is left regular, each left vertex has same degree. Therefore, by ignoring aforementioned edges (corresponding to $< 0.1|V|$ left vertices), we are ignoring atmost 0.1 fraction of the edges.

Now we give a labelling that covers much more than α fraction of the edges, thereby arriving at a contradiction (Since, in a NO instance, no labelling covers more than α fraction of edges)

Define the following labelling to V and W . The label of $w \in W$ is an arbitrary label from set T_w . The label of $v \in V$ is a random label from set T_v . We show that this labelling satisfies, on expectation, at least $0.9\alpha/0.1$ edges. Consider any arbitrary edge (v, w) such that $v \notin V^*$. Since $v \notin V^*$, we have $|T_v| \leq 0.1/\alpha$. Let label of w be some $i^* \in T_w$. By *Lemma (2.4)*, we know that there exists a label $j^* \in T_v$ such that $\pi_{v,w}(j^*) = i^*$. With probability $1/|T_v| > \alpha/0.1$, we select j^* as the label of v and the edge (v, w) is satisfied. (Since we are *randomly* selecting label of v from T_v , the probability of landing on right label is $1/|T_v|$. Further since $|T_v| < 0.1/\alpha$ for $v \notin V^*$, we get $1/|T_v| > \alpha/0.1$)

Thus, each edge is satisfied with probability $> \alpha/0.1$. There are $> 0.9|E|$ edges with left vertex not in V^* . Since, the label assignment corresponding to any edge doesn't care about other edges, the assignments are independent. Therefore, on expectation $> \frac{0.9\alpha}{0.1}E$ edges are satisfied, i.e., $> 9\alpha$ fraction of edges are satisfied, a contradiction. Therefore Case 2 cannot occur. This completes the proof.

Remark 29. In YES case, we need to show “there exists” an assignment that gives $OBJ \leq \sqrt{2mn}$. That is done by explicitly giving the assignment.

Remark 30. In NO case, we show that “for all” assignments to variables in Y , $OBJ \geq 0.1\sqrt{mn/\alpha}$.

2.1.5 Technical Analysis

Let n be the size of SAT instance. Combining *Thm(2.3)* with the Reduction from Label Cover to GapCVP (via *Problem X*), we get a reduction from SAT to GapCVP with $Gap = \frac{1}{20}R^{\beta/2}$ that runs in time $n^C \log R$ for some constant C .

Since we have freedom with choice of R , we choose $R = 2^{(\log n)^k}$ for some large enough integer k . (Note that we are shifting the freedom of choice from R to k) The size of CVP instance is obviously less than the time allowed for reduction. Let N denote the size of CVP instance. We have $N \leq n^C \log R$. Thus $\log N \leq C \log R \log n$. Substituting $R = 2^{(\log n)^k}$, we get:

$$\begin{aligned}
\log N &\leq C \cdot (\log n)^k \cdot \log n \\
&\leq C \cdot (\log n)^{k+1} \\
&\leq (\log n)^{k+2} \quad [\text{Since constant } C \leq \log n.] \\
&\quad \underline{\text{Constraint not tight}} \\
&\Rightarrow (\log N)^{1/(k+2)} \leq \log n
\end{aligned} \tag{8}$$

Now, consider the gap:

$$\begin{aligned}
Gap &= \frac{1}{20}R^{\beta/2} = \frac{1}{20}2^{(\beta/2) \cdot \log R} \\
&= \frac{1}{20}2^{(\beta/2) \cdot (\log n)^k} \\
&\geq \frac{1}{20}2^{(\log n)^{k-1}} \quad [\text{Since } \beta/2 \geq \log n] \quad \underline{\text{Constraint not tight}} \\
&\geq 2^{(\log N)^{(k-1)/(k+2)}}
\end{aligned} \tag{9}$$

Since we are choosing k as a large integer, consider $k \approx 3/\varepsilon$. We have $(k-1)/(k+2) = 1 - \frac{3}{k+2} \approx 1 - \varepsilon$. Thus, the gap is $\approx 2^{(\log N)^{(1-\varepsilon)}}$.

2.1.6 Rank Analysis

We can represent the affine constraints as a system of linear equations. Let A be the corresponding matrix we get. We can formulate the affine constraints (3)-(5) as $AX = Y$.

- Number of variables = $n \cdot R + m \cdot S$
- Number of equations = $n + m + |E| \cdot S$.
- Dimension of $A = (n + m + |E| \cdot S) \times (n \cdot R + m \cdot S)$
- Dimension of $X = (n \cdot R + m \cdot S) \times 1$
- Dimension of $Y = (n + m + |E| \cdot S) \times 1$
- Structure of Y : $[1^n \ 1^m \ 0^{|E| \cdot S}]^T$

Define $A|Y$ to be the augmented matrix. We have:

- If $\text{rank}(A) < \text{rank}(A|Y)$: No solution
- If $\text{rank}(A) = \text{rank}(A|Y) = \#variables$: Unique solution
- If $\text{rank}(A) = \text{rank}(A|Y) < \#variables$: Infinitely many solutions

In case of YES instance we will always have $\text{rank}(A) = \text{rank}(A|Y)$. In case of NO instance, we can have $\text{rank}(A) < \text{rank}(A|Y)$ or $\text{rank}(A) = \text{rank}(A|Y)$.

We have possibility of $\text{rank}(A) = \text{rank}(A|Y)$ in NO instance because NO instance only guarantees that we cannot have binary assignments to variables and satisfy all constraints. But, we can definitely have the case that all constraints (3)-(5) are satisfied when we allow non-binary integer values to variables.

Remark 31. $\text{rank}(A) \leq n \cdot R + m \cdot S$

Here n = number of left vertices, m = number of right vertices, R = number of left labels, S = number of right labels.

3 Projection Games

Input: Bipartite graph $G = (V, W, E)$; finite alphabets $[R], [S]$; constraints (or projections) $\pi_e : [R] \rightarrow [S]$ for every edge $e \in E$.

The goal is to find assignments to vertices $\phi_V : V \rightarrow [R]$, $\phi_W : W \rightarrow [S]$ that satisfy as many edges as possible. We say that an edge $e = (v, w) \in E$ is satisfied, if the projection constraint holds i.e., $\pi_e(\phi_V(v)) = \phi_W(w)$. We say that G is a *projection game* if every constraint π that appears in the game is a projection constraint, i.e., for every edge $e \in E$, each $\beta \in [S]$ has atmost one $\alpha \in [R]$ for which $(\alpha, \beta) \in \pi_e$.

We denote the size of projection game by $n = |V| + |W| + |E|$. The size of the alphabet of projection game is $\max\{R, S\}$

4 Projection Games Conjecture

4.1 Known results (without PGC)

Theorem 4.1. [MR10] *There exists a constant $\beta > 0$ such that for any instance of SAT of size n , there exists an efficient reduction to a projection game G with the following properties:*

- The size of the projection game, denoted $N = \text{size}(G)$, is $n^{1+o(1)}$.
- The soundness error of this projection game is $\varepsilon = 1/(\log N)^\beta$.

Theorem 4.2. [MR10] *There exists $c > 0$, such that for every $\varepsilon \geq 1/N^c$, SAT on input of size n can be reduced to a projection game of size $N = n^{1+o(1)} \cdot \text{poly}(1/\varepsilon)$. The projection game is over an alphabet of size exponential in $1/\varepsilon$, and has soundness error ε . The reduction can be computed in linear time in the size and the alphabet size of the projection game. The projection game is a bi-regular graph whose degrees are $\text{poly}(1/\varepsilon)$.*

4.2 Conjecture

Conjecture 4.3. [Mos12] PGC: *There exists $c > 0$, such that for every $\varepsilon \geq 1/N^c$, SAT on input of size n can be efficiently reduced to a projection game of size $N = n^{1+o(1)} \text{poly}(1/\varepsilon)$ over an alphabet of size $\text{poly}(1/\varepsilon)$ that has soundness error ε .*

In essence, the conjecture states that there exists $c > 0$ such that it is NP-hard to distinguish, given a projection game G of size N , between the case when $\text{value}(G) = 1$ and $\text{value}(G) < \varepsilon$, where $\varepsilon \geq 1/N^c$. In other words, there exists $c > 0$ such that it is NP-hard to approximate a projection game with size N , and alphabet size $\text{poly}(N)$, to a factor of N^c .

We can assume without loss of generality that the projection game in Conjecture 4.3 is *bi-regular*. We note that any projection game can be converted to bi-regular using a technique developed in [MR10], and the cost in the soundness error and graph size does not change the parameters stated in Conjecture 4.3. Furthermore, we can assume the left and the right label sets to be the same. (Take union of both label sets as the new common label set)

4.3 Technical Analysis

We have $\varepsilon \geq 1/N^c$. Let $\varepsilon = 1/N^{c-\alpha}$ for some $\alpha > 0$.

$$\begin{aligned} \text{We have } N &= n^{1+o(1)} \cdot \text{poly}(N^{c-\alpha}) \\ \Rightarrow N &= n^{1+o(1)} \cdot N^{c_1(c-\alpha)} \\ \Rightarrow N &= n^{\frac{1+o(1)}{1-c_1(c-\alpha)}} \end{aligned} \tag{10}$$

Furthermore the alphabet size of projection game is $\text{poly}(1/\varepsilon) = N^{c_2(c-\alpha)}$.

We have soundness error ε . Restating it in terms of n , we get $\varepsilon = N^{-(c-\alpha)} = n^{-\frac{(1+o(1)) \cdot (c-\alpha)}{1-c_1(c-\alpha)}}$

4.4 PG $\rightarrow$ CVP

Recall that we have a reduction from PG to CVP that gives a CVP instance with $\text{rank} = n \cdot R + m \cdot S$. In our context, $\text{rank} = n \cdot R + m \cdot S \leq 2 \cdot N \cdot N^{c_2(c-\alpha)}$. Thus, the rank of CVP instance as per PGC is:

$$\log(\text{rank}) \leq \left\lceil \frac{(1+o(1)) \cdot (1+c_2(c-\alpha))}{1-c_1(c-\alpha)} \right\rceil \cdot \log n + \log 2 \tag{11}$$

Since both the product terms in the numerator are greater than 1, and the denominator is less than 1, the fraction is greater than 1. Thus, we can restate the bound on rank as:

$$\text{rank} \leq 2 \cdot n^{(1+o(1)) \cdot (1+\beta)}$$

Here, $(1+\beta) = \frac{1+c_2(c-\alpha)}{1-c_1(c-\alpha)}$

4.4.1 Conclusion

- Assuming PGC, we can efficiently reduce SAT instance on input of size n to CVP instance with rank atmost $\text{poly}(n)$. We are unable to get rank quadratic or linear in n , by just assuming PGC.
- Assuming PGC, there exists $c > 0$, such that approximating CVP to within N^c on an N -dimensional lattice is NP-hard.
- Without assuming PGC, the best we have is:

Theorem 4.4. [Kho09] *For any constant $\varepsilon > 0$, there is a reduction from SAT instance ϕ of size n to GapCVP instance with gap $2^{(\log N)^{1-\varepsilon}}$. The reduction runs in time $N \leq n^{C \log R}$ for some constant C .* **2.1.5**

Note that the reduction is not poly-time. Here, $R = 2^{(\log n)^{3/\varepsilon}}$. Thus, the reduction runs in time $n^{C \log R} = n^{(\log n)^{3/\varepsilon}}$. The rank we get here is $n \cdot R + m \cdot S = O(n \cdot 2^{(\log n)^{3/\varepsilon}})$.

5 Proposal

Restating known results:

- **Theorem 4.2**[MR10] There exists $c > 0$, such that for every $\varepsilon \geq 1/N^c$, SAT on input of size n can be reduced to a projection game of size $N = n^{1+o(1)} \cdot \text{poly}(1/\varepsilon)$. The projection game is over an alphabet of size exponential in $1/\varepsilon$, and has soundness error ε . The reduction can be computed in linear time in the size and the alphabet size of the projection game. The projection game is a bi-regular graph whose degrees are $\text{poly}(1/\varepsilon)$.
- **PGC** [Mos12] There exists $c > 0$, such that for every $\varepsilon \geq 1/N^c$, SAT on input of size n can be efficiently reduced to a projection game of size $N = n^{1+o(1)} \text{poly}(1/\varepsilon)$ over an alphabet of size $\text{poly}(1/\varepsilon)$ that has soundness error ε .

5.1 Proposed Conjecture

5.1.1 Conjecture

Conjecture 5.1. *There exists $c > 0$, such that for every $\varepsilon \geq 1/N^c$, SAT on input of size n can be reduced to a projection game of size $N = n^{1+o(1)} \text{poly}(1/\varepsilon)$ over an alphabet of size $2^{\text{polylog}(1/\varepsilon)}$ that has soundness error ε . The reduction can be computed in linear time in the size and the alphabet size of the projection game, i.e., $2^{\text{polylog}(1/\varepsilon)}$ time.*

5.1.2 Analysis

We have $\varepsilon \geq 1/N^c$. Let $\varepsilon = 1/N^{c-\alpha}$ for some $\alpha > 0$.

$$\begin{aligned}
 \text{We have } N &= n^{1+o(1)} \cdot \text{poly}(N^{c-\alpha}) \\
 &\Rightarrow N = n^{1+o(1)} \cdot N^{c_1(c-\alpha)} \\
 &\Rightarrow N = n^{\frac{1+o(1)}{1-c_1(c-\alpha)}}
 \end{aligned} \tag{12}$$

Furthermore the alphabet size of projection game is:

$$\begin{aligned}
& 2^{\text{polylog}(1/\varepsilon)} \\
&= 2^{\lceil \log(N^{c-\alpha}) \rceil^{c_2}} \\
&= 2^{(c-\alpha)^{c_2} \cdot (\log N)^{c_2}} \\
&\text{Let } \beta = \frac{1 + o(1)}{1 - c_1(c - \alpha)} \\
&\quad \Rightarrow N = n^\beta \\
&\text{Let } \lambda = 2^{(c-\alpha)^{c_2}} \tag{13} \\
&\quad \Rightarrow \text{Alphabet size} = \lambda^{(\log N)^{c_2}} \\
&= \lambda^{\beta^{c_2} \cdot (\log n)^{c_2}} = \lambda_1^{(\log n)^{c_2}} \quad [\lambda_1 = \lambda^{\beta^{c_2}}] \\
&\quad \Rightarrow \text{Alphabet size} = O(2^{\text{polylog}(n)}) \\
&\text{Using the PG} \rightarrow \text{CVP, we get rank} \leq O(N \cdot \text{Alphabet size}) \\
&\quad \text{rank} \leq O(n^\beta \cdot 2^{\text{polylog}(n)})
\end{aligned}$$

5.1.3 Reasoning

- The bound on CVP rank we get from the proposed conjecture 5.1 is in agreement with the best known bounds on rank achieved without assuming PGC. 2.1.5
- Theorem 4.2, and PGC 4.3 leave room for super-polynomial time reductions. Proposed conjecture 5.1 addresses this particular case.
- The structure of proposed conjecture is on the same lines as Theorem 4.2, and PGC 4.3.
- The CVP rank bound obtained from the proposed conjecture 5.1 further strengthens the confidence in the conjecture.

References

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